ONLINE APPENDIX A

No Data Left Behind – A Primer on Bayesian State-Space Models for Acceptance Sampling

Henrik R. Godmann[1,*], Lotte Mensink[2],
Ruud Wetzels[2,3], & Eric-Jan Wagenmakers[1]

1-University of Amsterdam
2-Nyenrode Business University
3-PwC Advisory

*Correspondence concerning this appendix or the main manuscript should be addressed to Henrik R. Godmann, Department of Psychological Methods, University of Amsterdam, Nieuwe Achtergracht 129, Amsterdam, The Netherlands. E-Mail: hrgodmann@gmail.com / h.r.godmann@uva.nl.

1 Lot-by-Lot Results

This appendix shows the lot-by-lot results (the evolving posterior distributions and predictive distributions) for every data scenario of the running example.

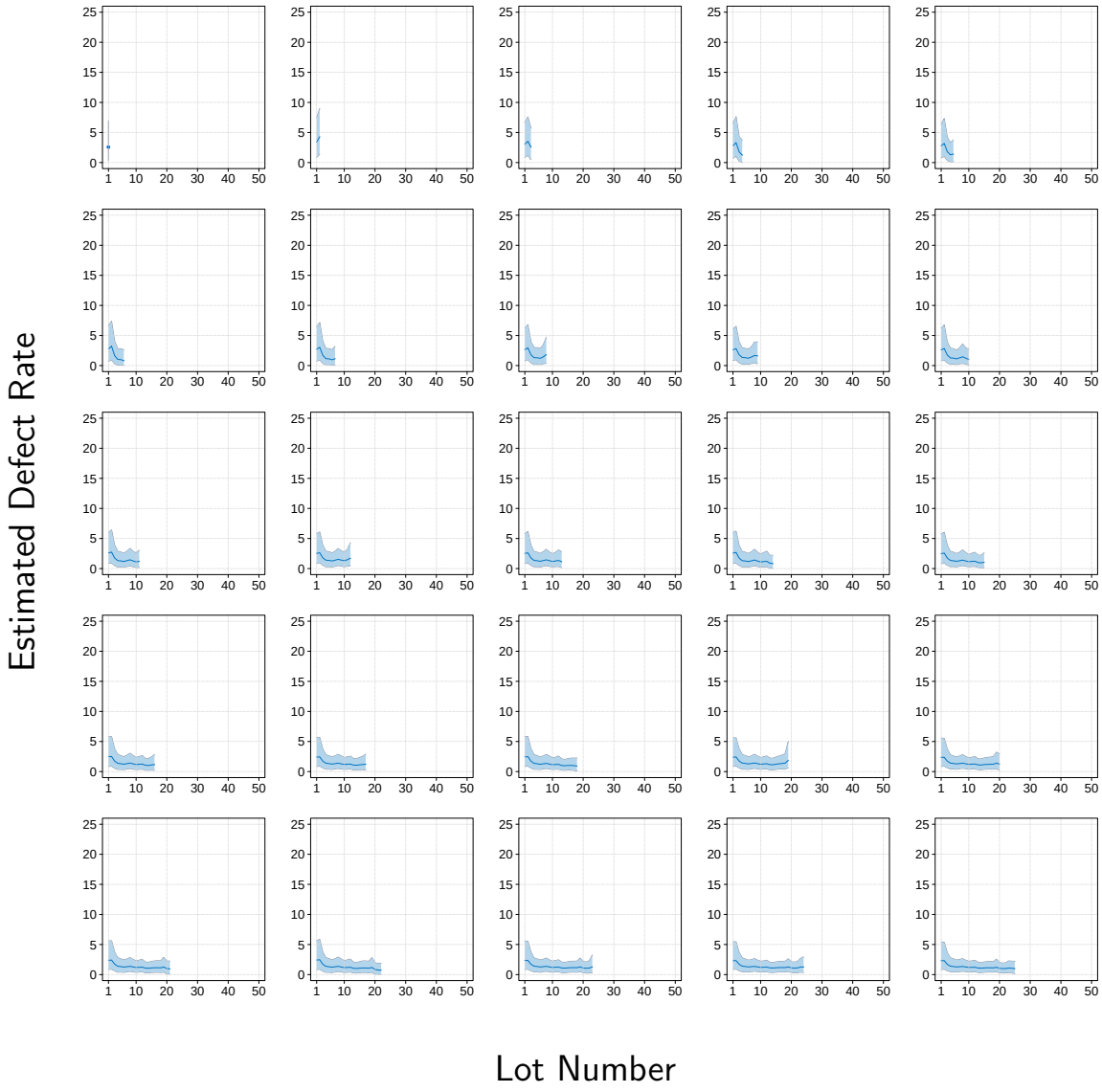


Figure 1: PART I, Stable good quality: Evolving posterior distributions of θ at each time point based on the Bayesian state-space model and the samples reused from the curtailed frequentist run. Each panel represents one day, starting from day 1 (Lot 1) until day 25 (of a total of 50 lots). The x-axis represents the lots from day 1 to n for each iteration, and the y-axis represents the estimated defect rate.

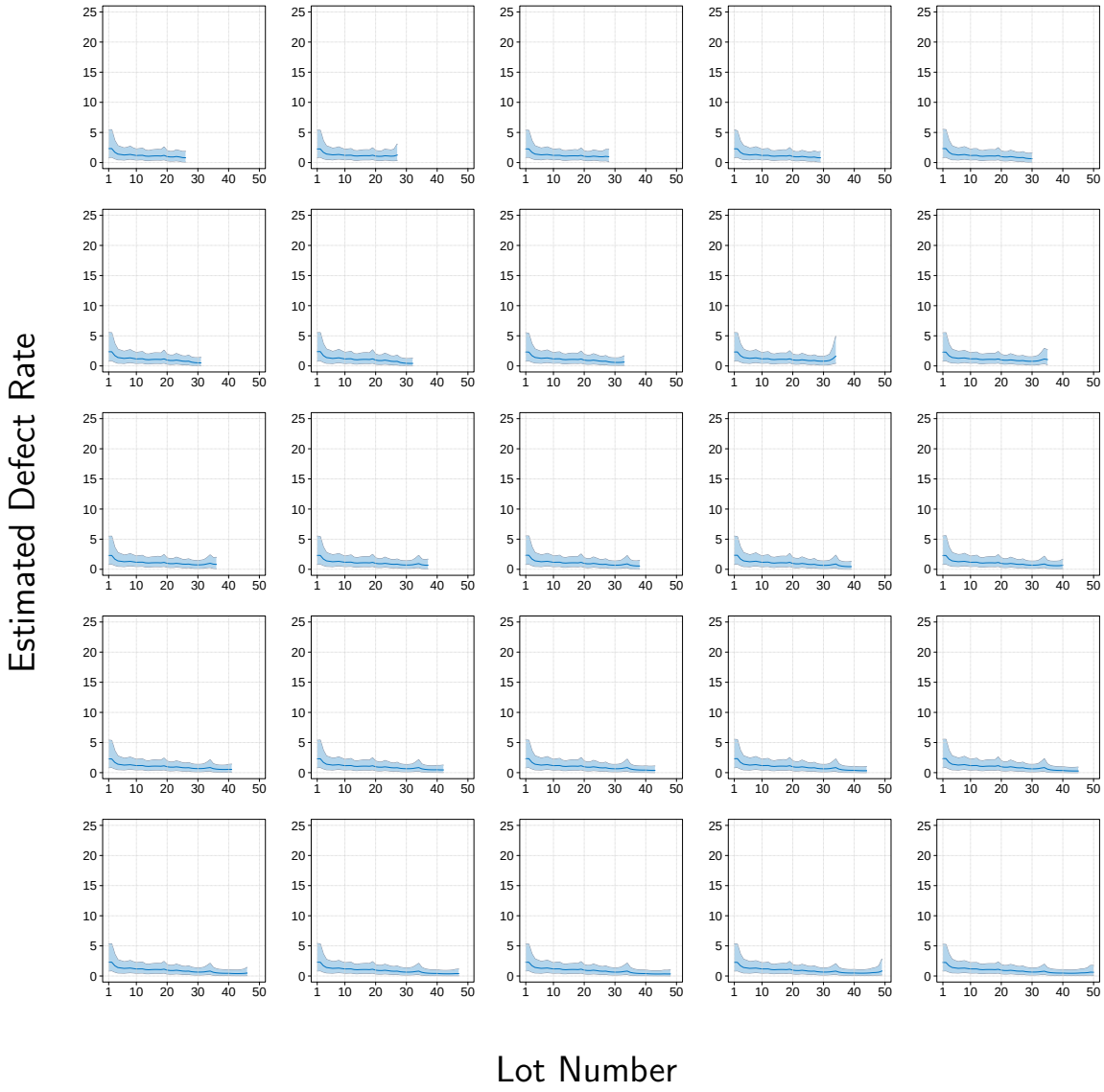


Figure 2: PART II, Stable good quality: Evolving posterior distributions of θ at each time point based on the Bayesian state-space model and the samples reused from the curtailed frequentist run. Each panel represents one day, starting from day 26 (Lot 26) until day 50 (of a total of 50 lots). The x-axis represents the lots from day 1 to n for each iteration, and the y-axis represents the estimated defect rate.

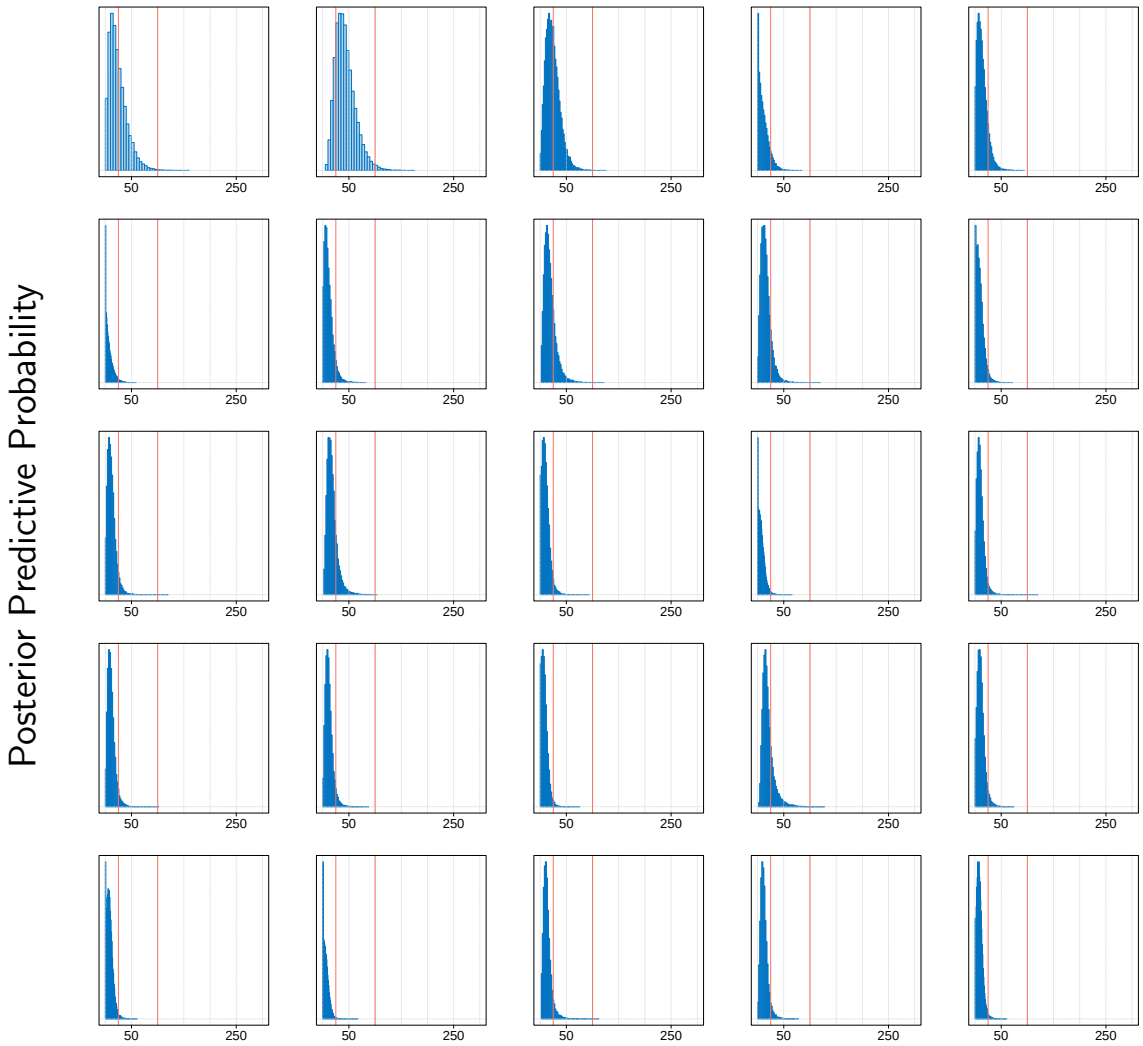


Figure 3: PART III, Stable good quality: Predictive distributions for the total number of defective items in the first 25 (out of 50 lots) lots based on the Bayesian state-space model and the samples reused from the curtailed frequentist run.

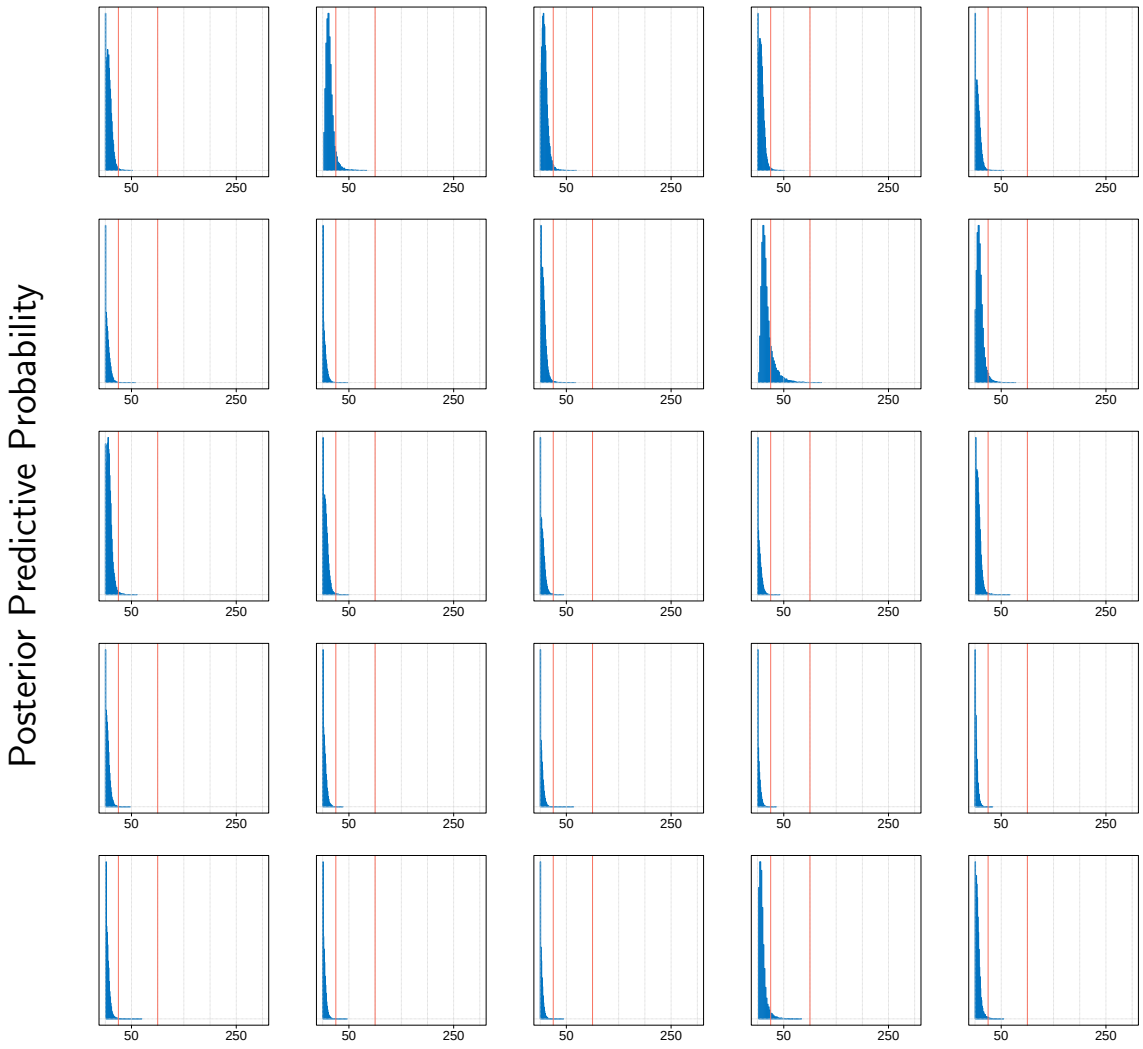


Figure 4: PART IV, Stable good quality: Predictive distributions for the total number of defective items in the 26th to the 50th lot based on the Bayesian state-space model and the samples reused from the curtailed frequentist run.

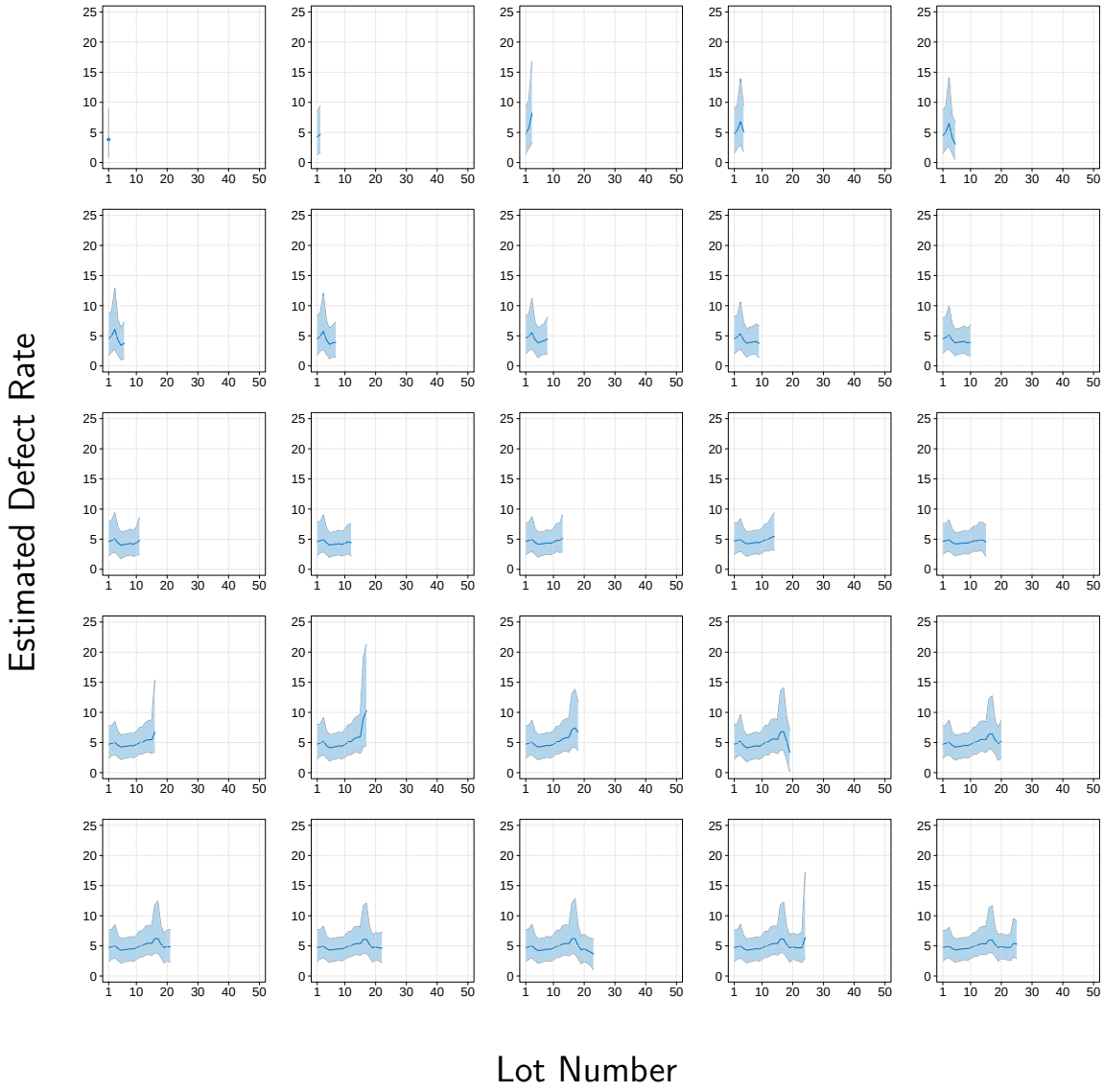


Figure 5: PART I, Stable intermediate quality: Evolving posterior distributions of θ at each time point based on the Bayesian state-space model and the samples reused from the curtailed frequentist run. Each panel represents one day, starting from day 1 (Lot 1) until day 25 (of a total of 50 lots). The x-axis represents the lots from day 1 to n for each iteration, and the y-axis represents the estimated defect rate.

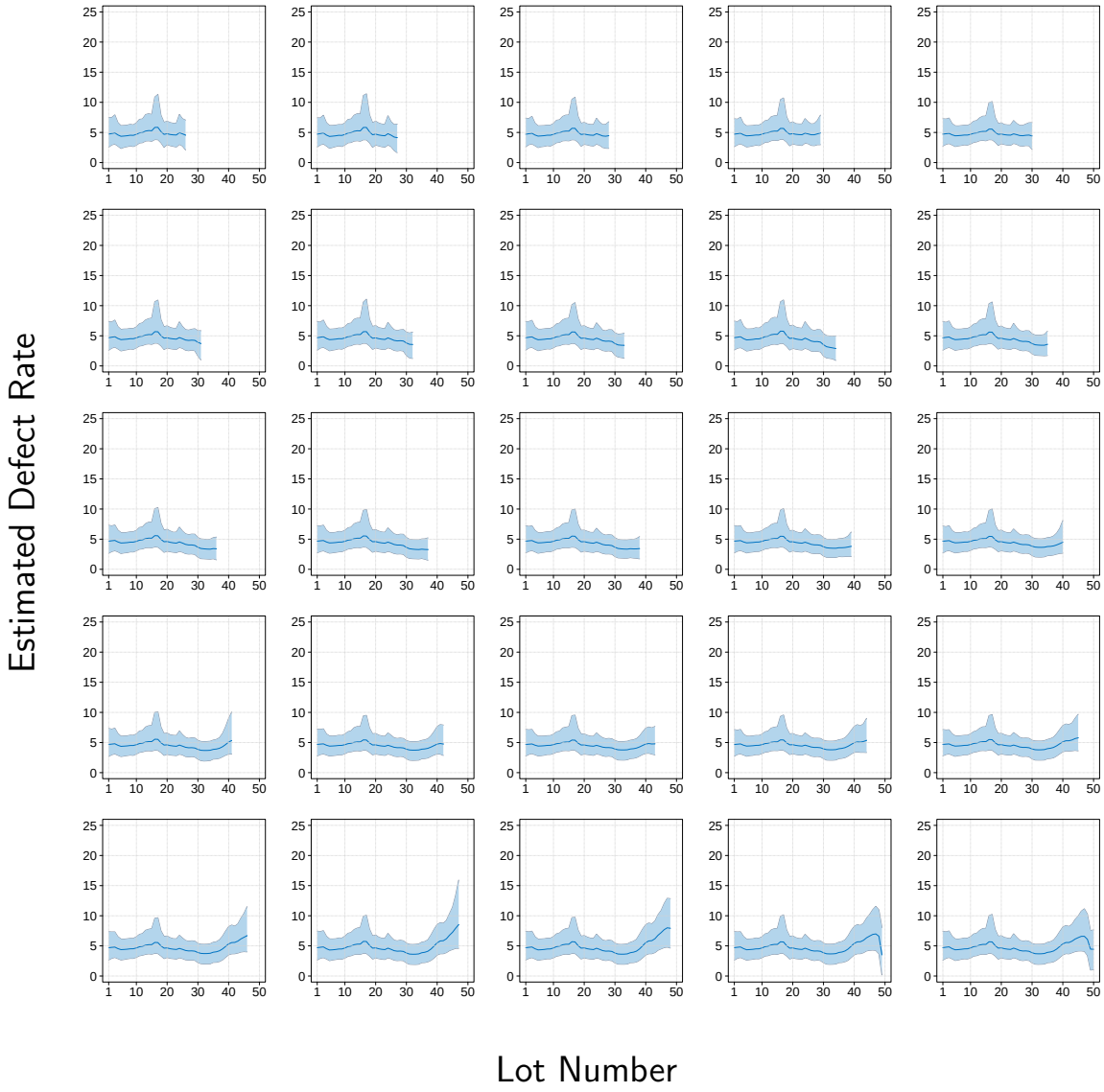


Figure 6: PART II, Stable intermediate quality: Evolving posterior distributions of θ at each time point based on the Bayesian state-space model and the samples reused from the curtailed frequentist run. Each panel represents one day, starting from day 26 (Lot 26) until day 50 (of a total of 50 lots). The x-axis represents the lots from day 1 to n for each iteration, and the y-axis represents the estimated defect rate.

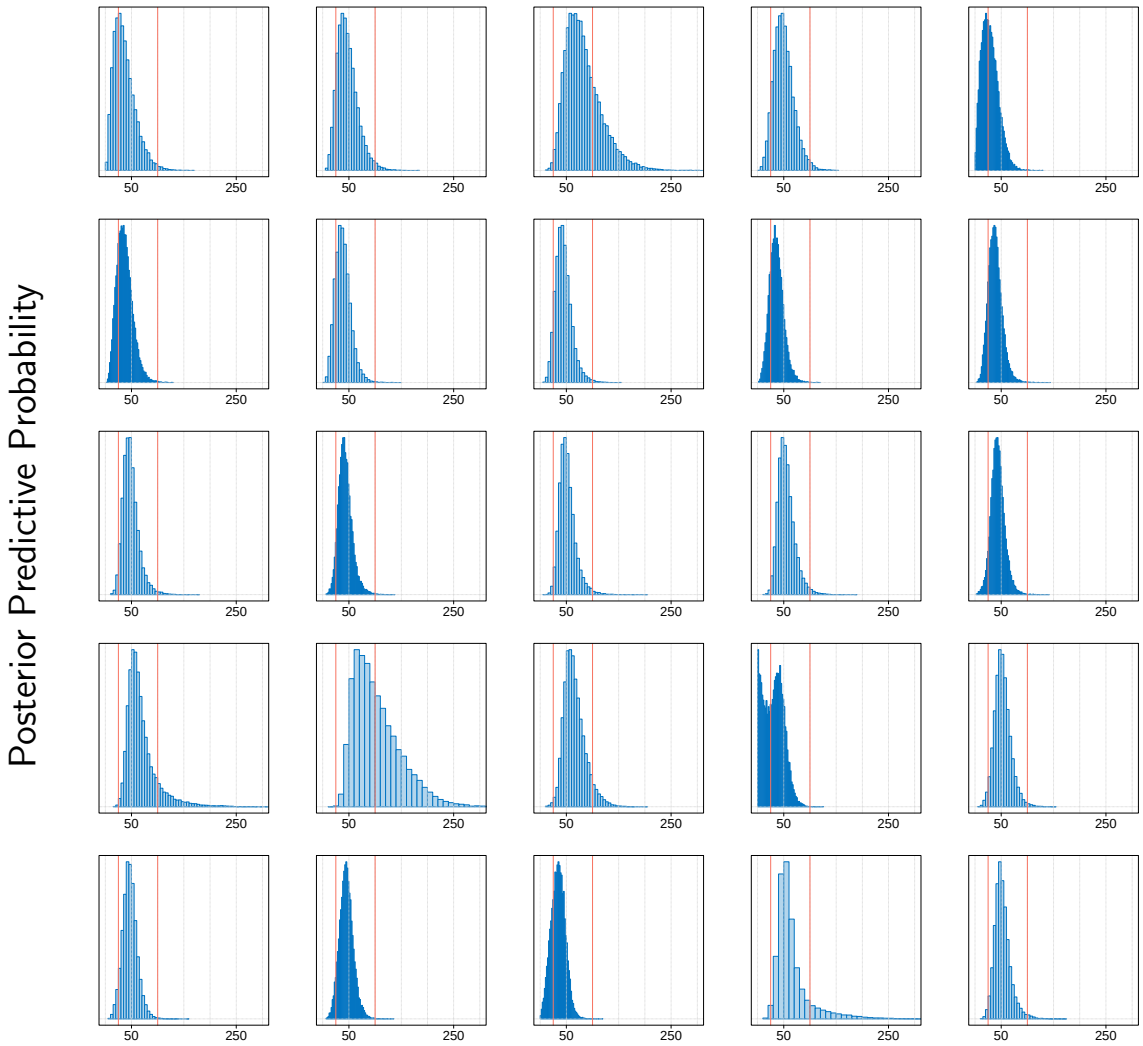


Figure 7: PART III, Stable intermediate quality: Predictive distributions for the total number of defective items in the first 25 (out of 50 lots) lots based on the Bayesian state-space model and the samples reused from the curtailed frequentist run.

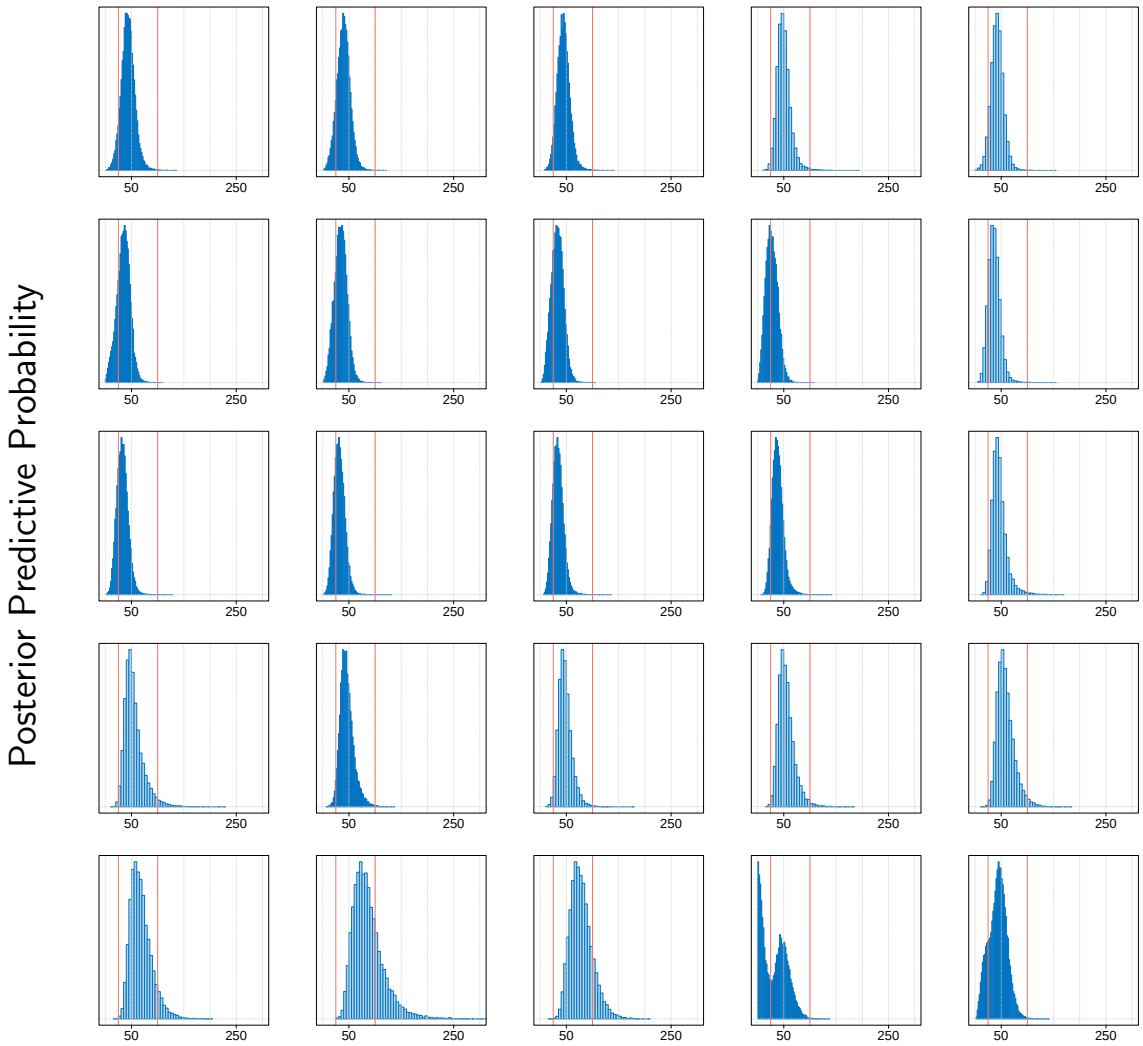


Figure 8: PART IV, Stable intermediate quality: Predictive distributions for the total number of defective items in the 26th to the 50th lot based on the Bayesian state-space model and the samples reused from the curtailed frequentist run.

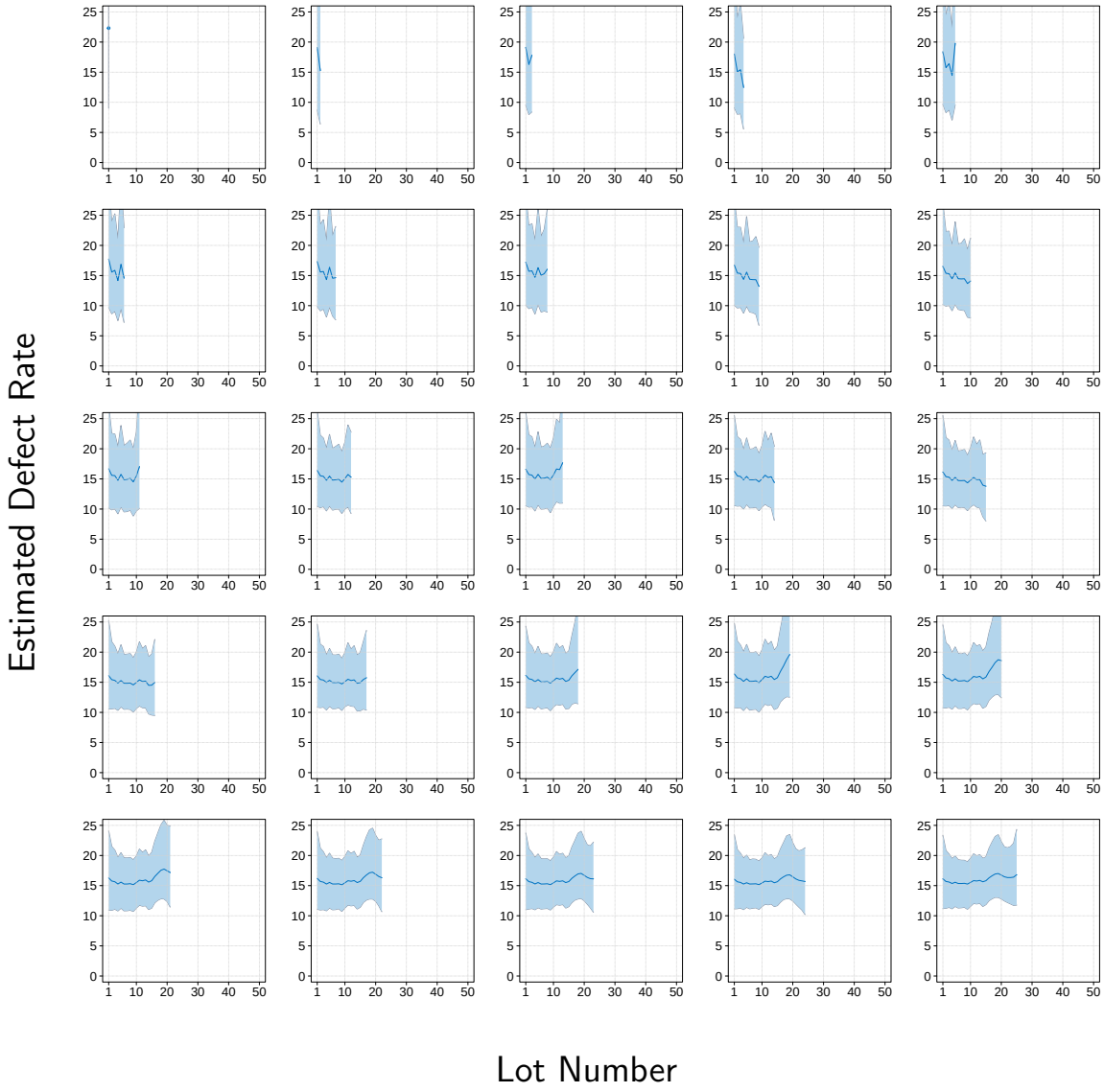


Figure 9: PART I, Stable poor quality: Evolving posterior distributions of θ at each time point based on the Bayesian state-space model and the samples reused from the curtailed frequentist run. Each panel represents one day, starting from day 1 (Lot 1) until day 25 (of a total of 50 lots). The x-axis represents the lots from day 1 to n for each iteration, and the y-axis represents the estimated defect rate.

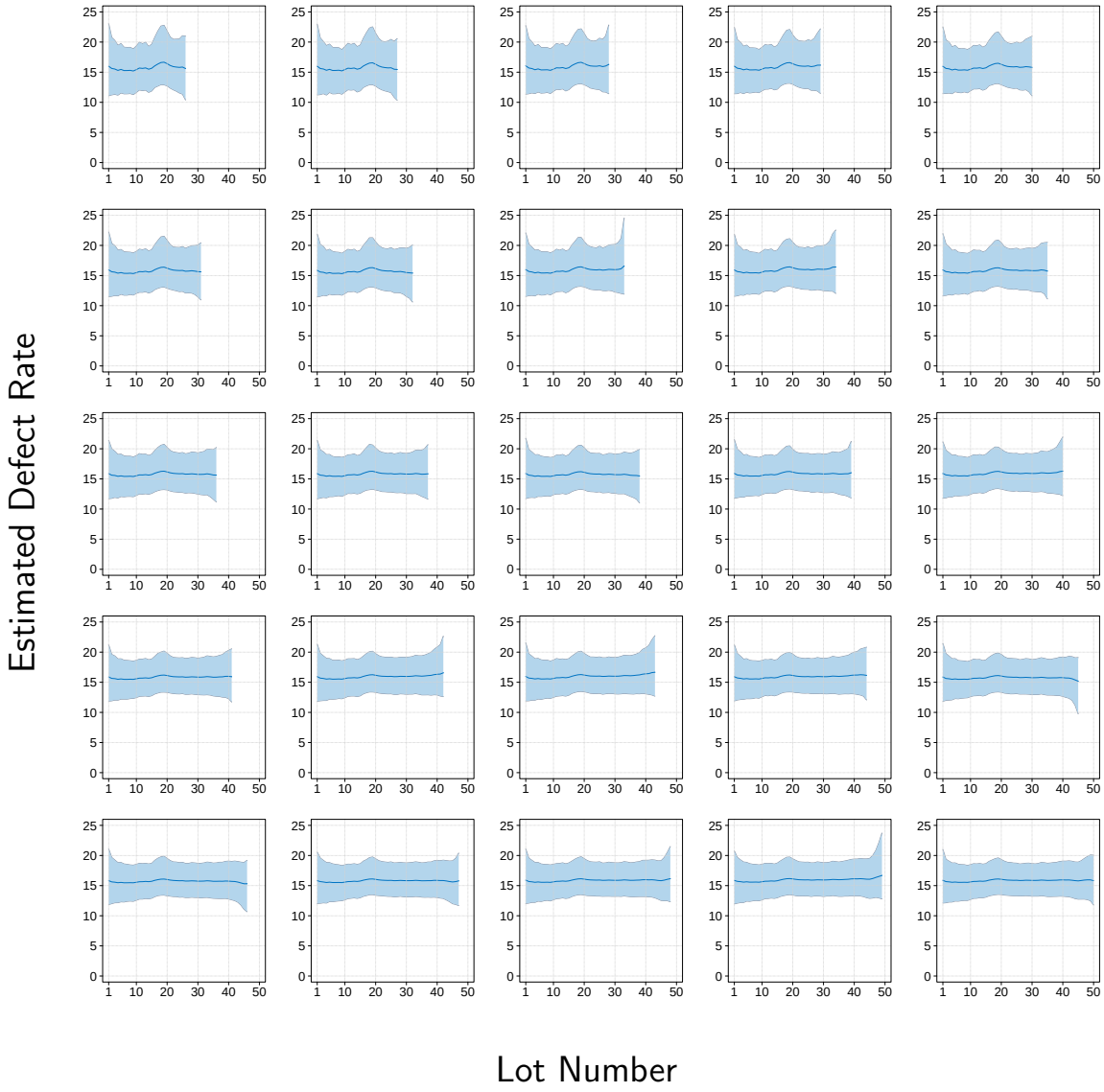


Figure 10: PART II, Stable poor quality: Evolving posterior distributions of θ at each time point based on the Bayesian state-space model and the samples reused from the curtailed frequentist run. Each panel represents one day, starting from day 26 (Lot 26) until day 50 (of a total of 50 lots). The x-axis represents the lots from day 1 to n for each iteration, and the y-axis represents the estimated defect rate.

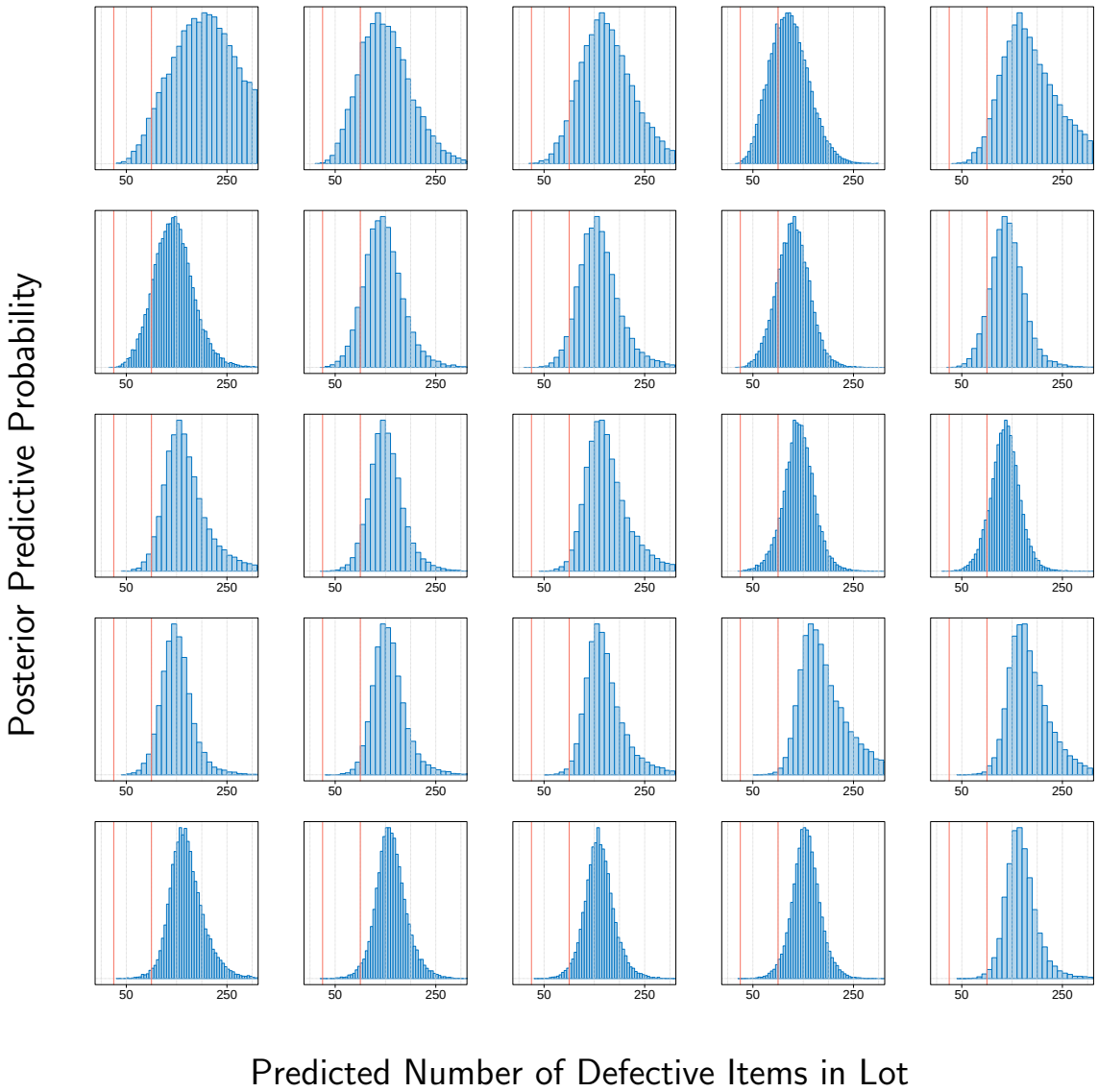


Figure 11: PART III, Stable poor quality: Predictive distributions for the total number of defective items in the first 25 (out of 50 lots) lots based on the Bayesian state-space model and the samples reused from the curtailed frequentist run.

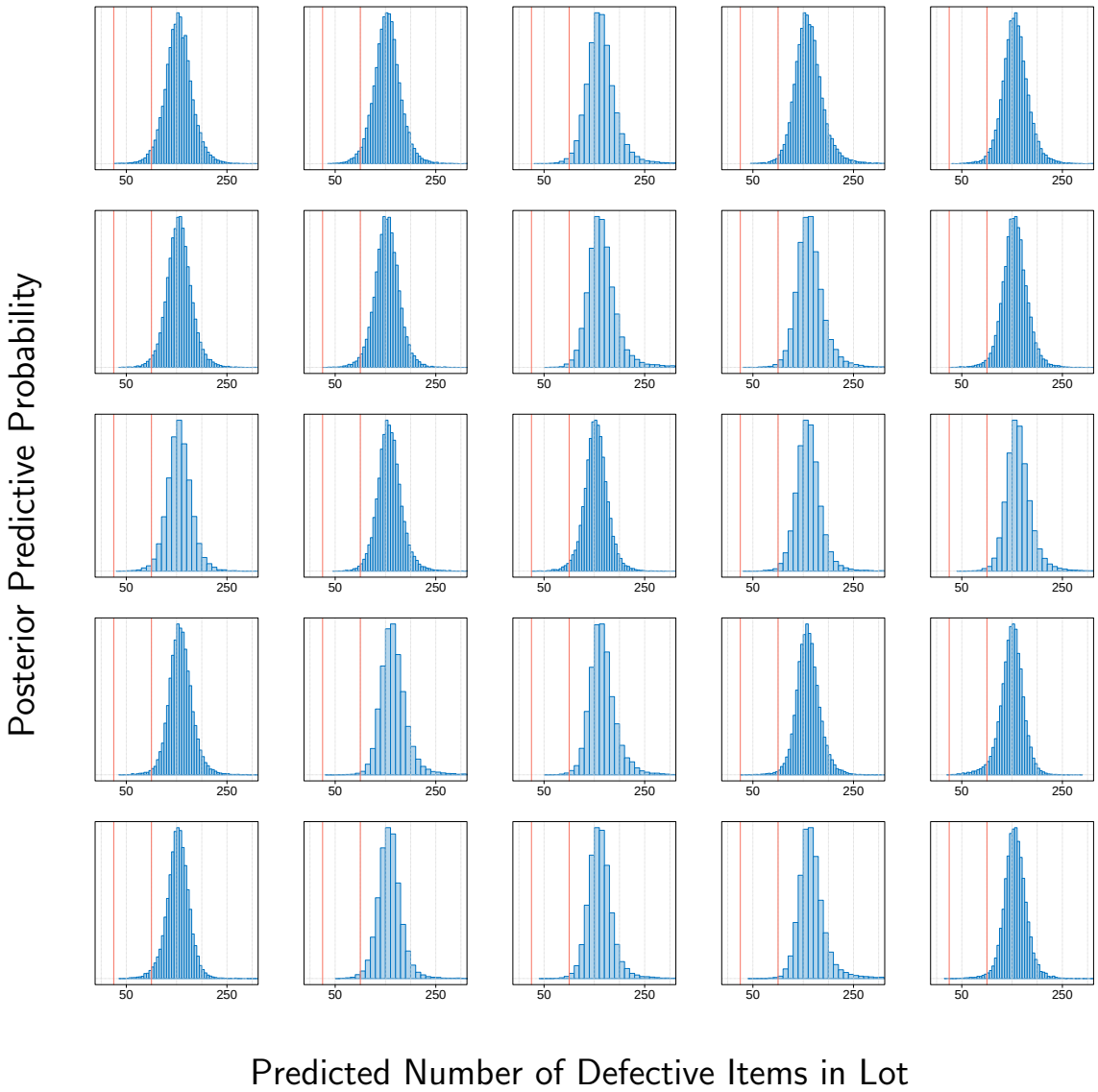


Figure 12: PART IV, Stable poor quality: Predictive distributions for the total number of defective items in the 26th to the 50th lot based on the Bayesian state-space model and the samples reused from the curtailed frequentist run.

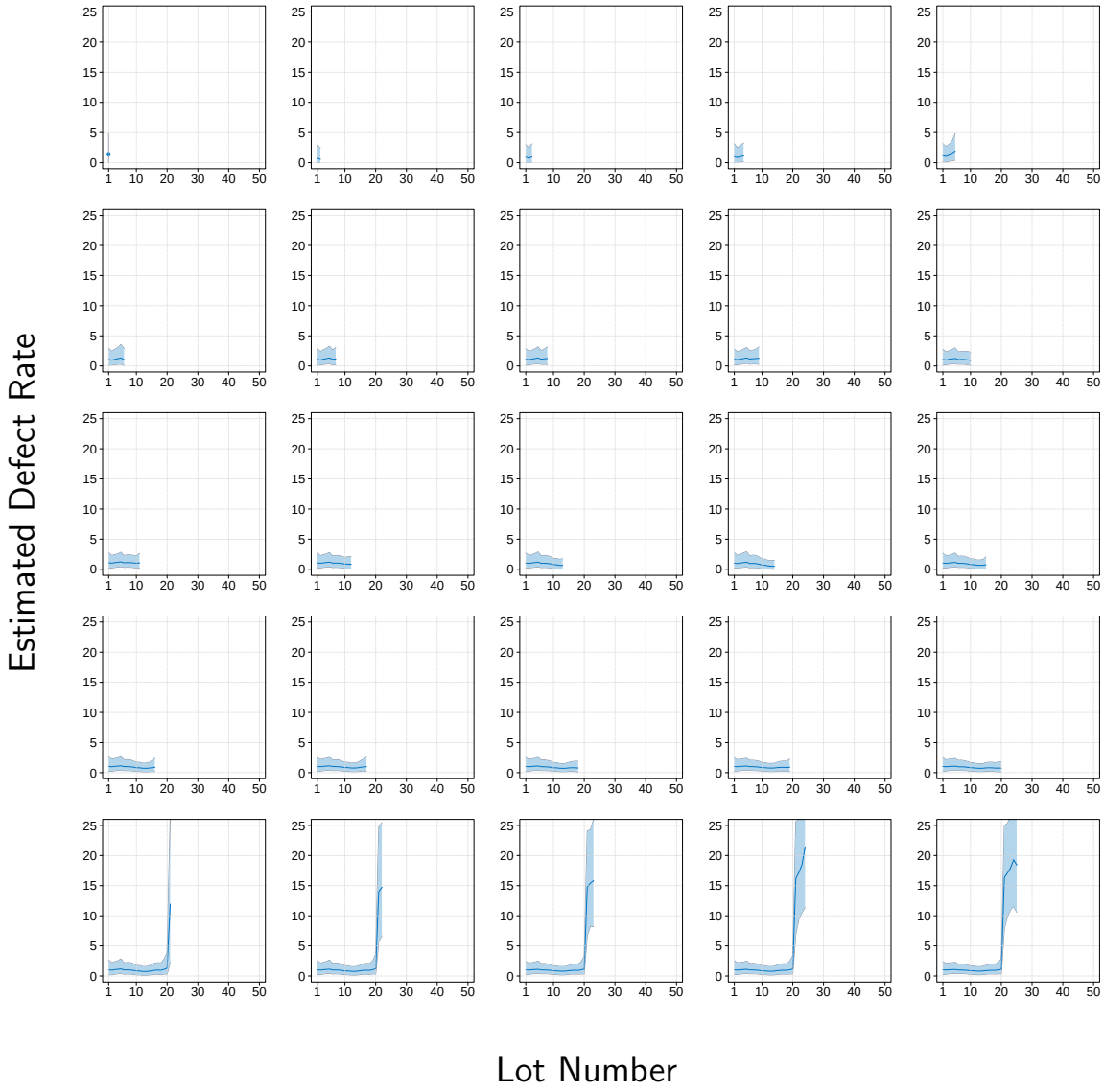


Figure 13: PART I, Abrupt, sustained deterioration: Evolving posterior distributions of θ at each time point based on the Bayesian state-space model and the samples reused from the curtailed frequentist run. Each panel represents one day, starting from day 1 (Lot 1) until day 25 (of a total of 50 lots). The x-axis represents the lots from day 1 to n for each iteration, and the y-axis represents the estimated defect rate.

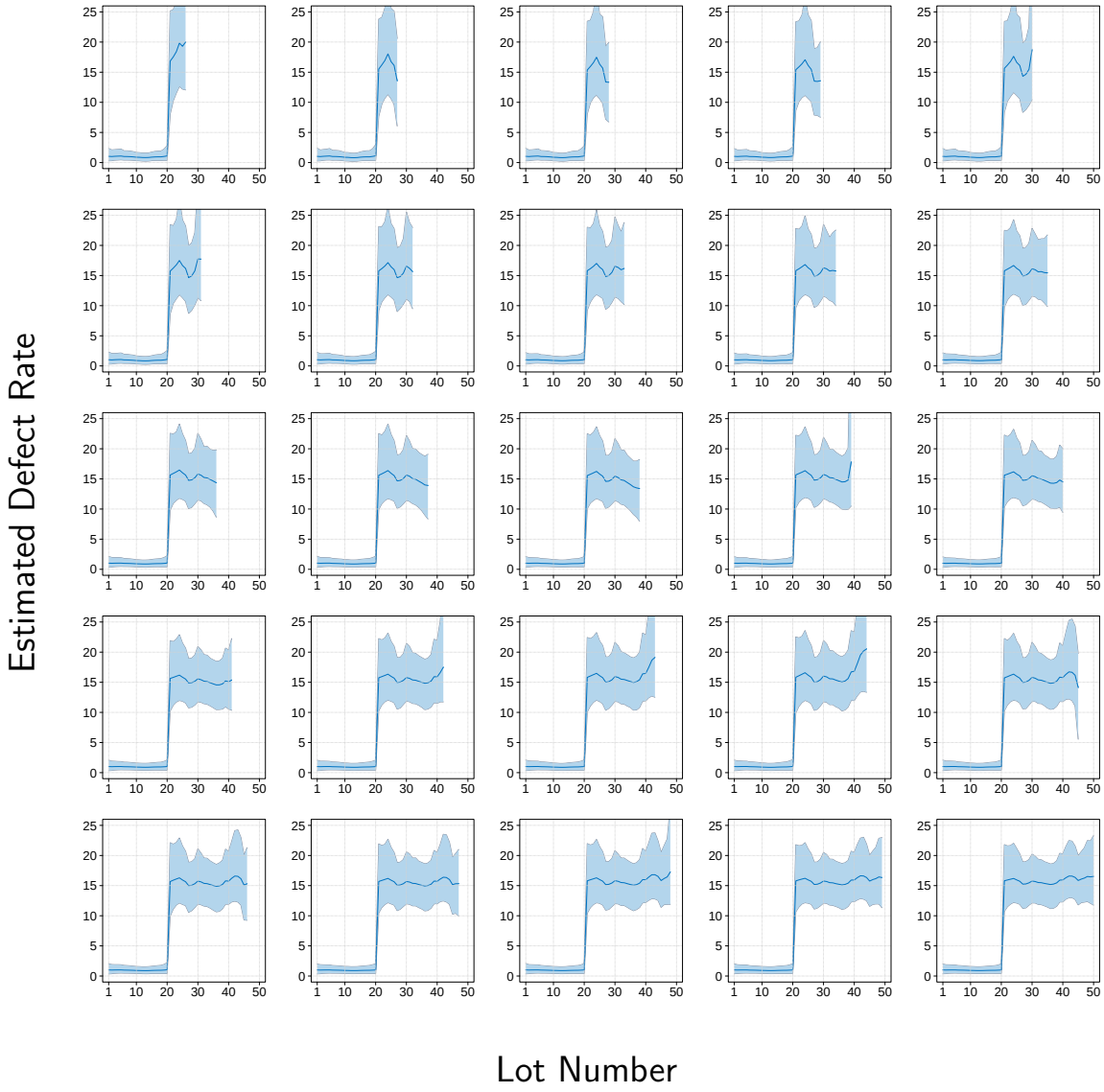


Figure 14: PART II, Abrupt, sustained deterioration: Evolving posterior distributions of θ at each time point based on the Bayesian state-space model and the samples reused from the curtailed frequentist run. Each panel represents one day, starting from day 26 (Lot 26) until day 50 (of a total of 50 lots). The x-axis represents the lots from day 1 to n for each iteration, and the y-axis represents the estimated defect rate.

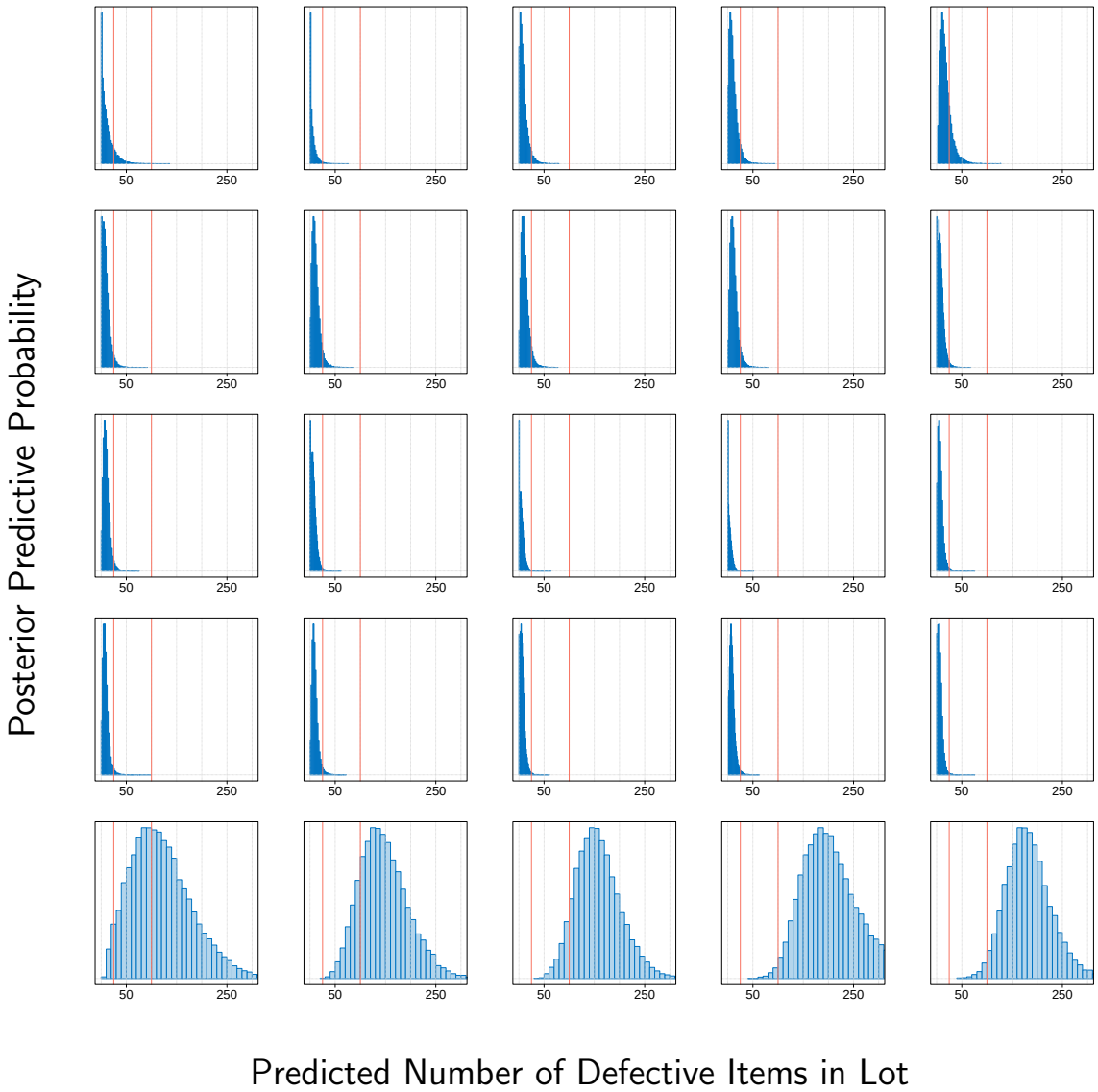


Figure 15: PART III, Abrupt, sustained deterioration: Predictive distributions for the total number of defective items in the first 25 (out of 50 lots) lots based on the Bayesian state-space model and the samples reused from the curtailed frequentist run.

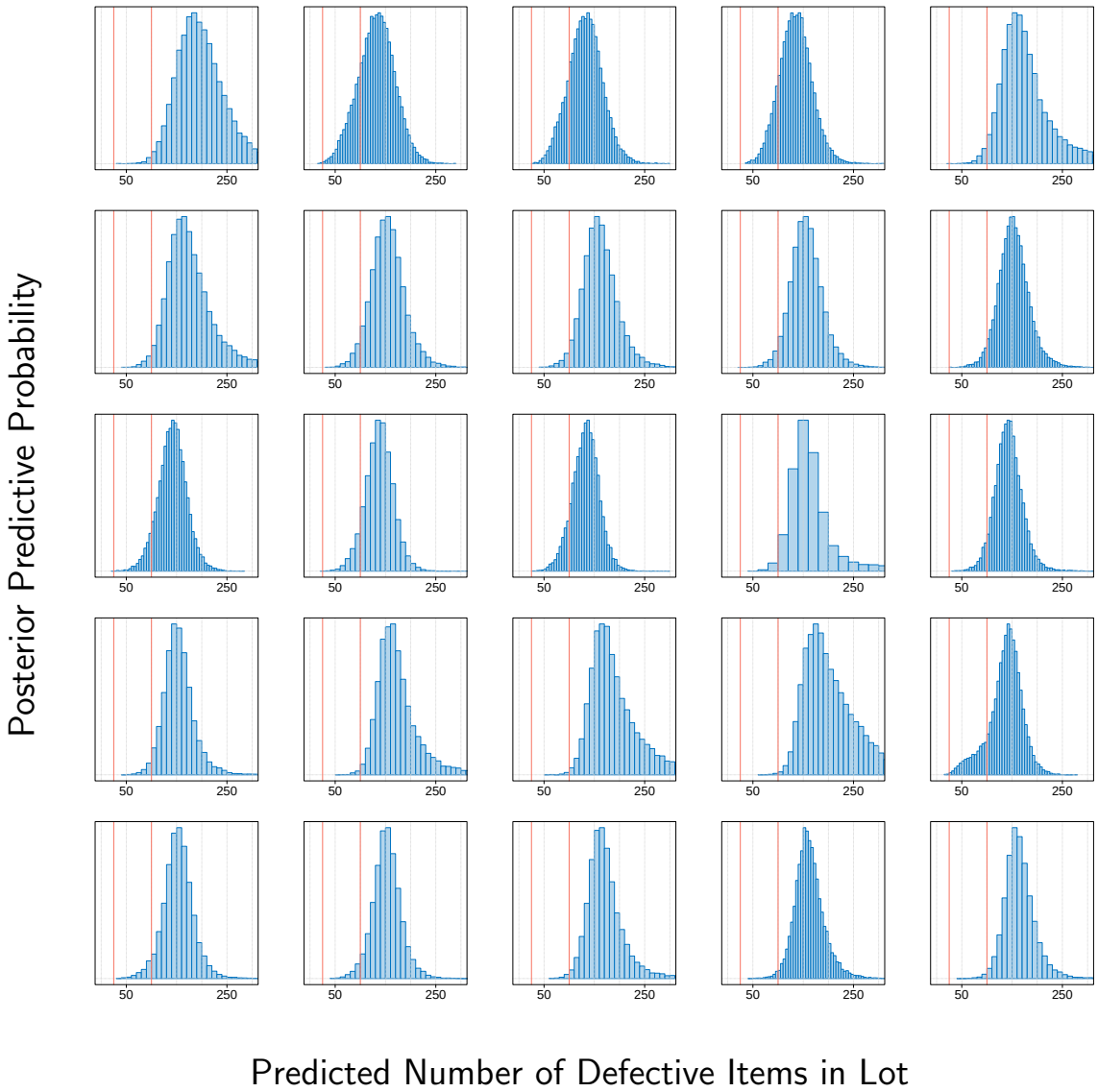


Figure 16: PART IV, Abrupt, sustained deterioration: Predictive distributions for the total number of defective items in the 26th to the 50th lot based on the Bayesian state-space model and the samples reused from the curtailed frequentist run.

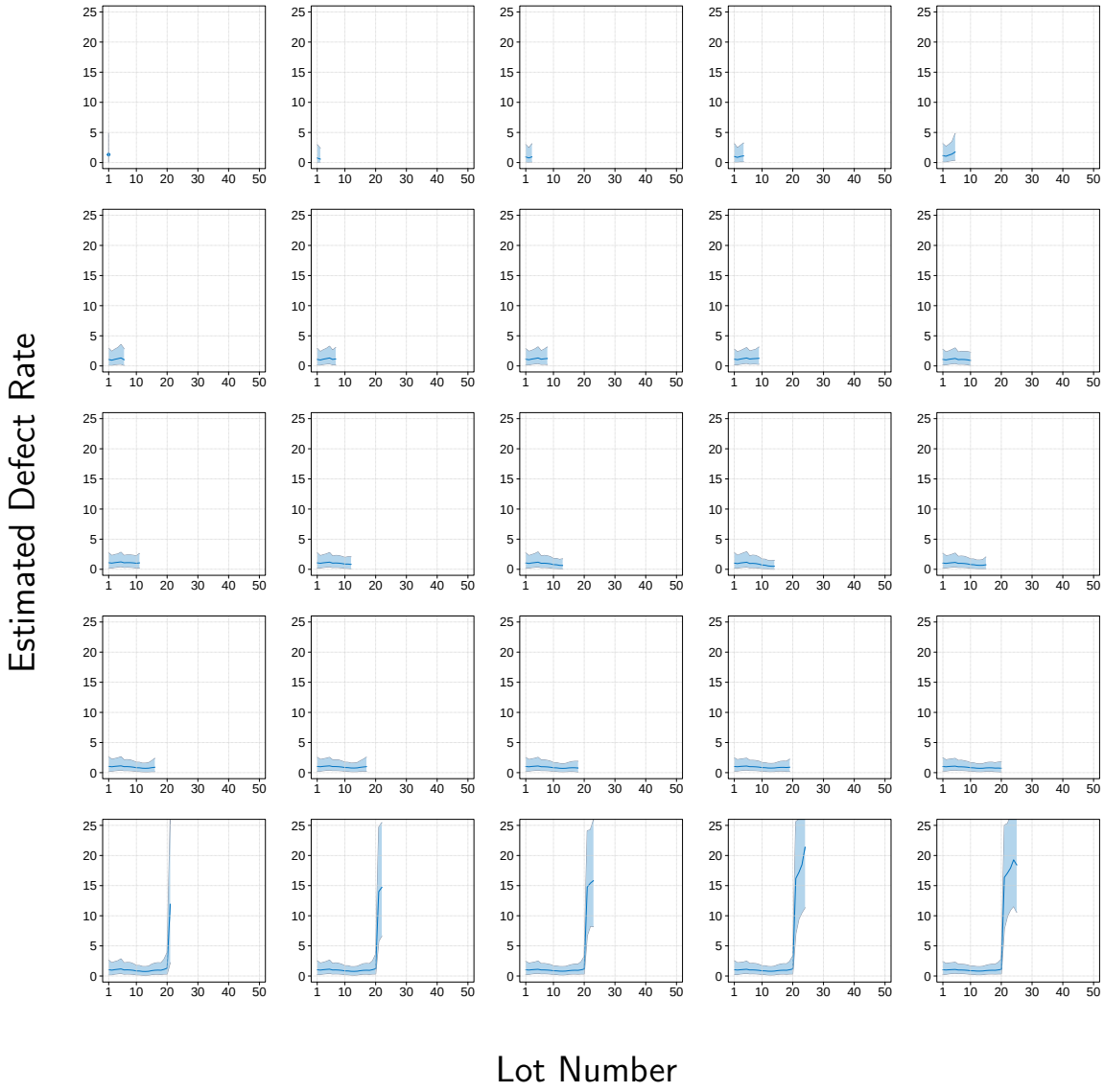


Figure 17: PART I, Abrupt deterioration and recovery: Evolving posterior distributions of θ at each time point based on the Bayesian state-space model and the samples reused from the curtailed frequentist run. Each panel represents one day, starting from day 1 (Lot 1) until day 25 (of a total of 50 lots). The x-axis represents the lots from day 1 to n for each iteration, and the y-axis represents the estimated defect rate.

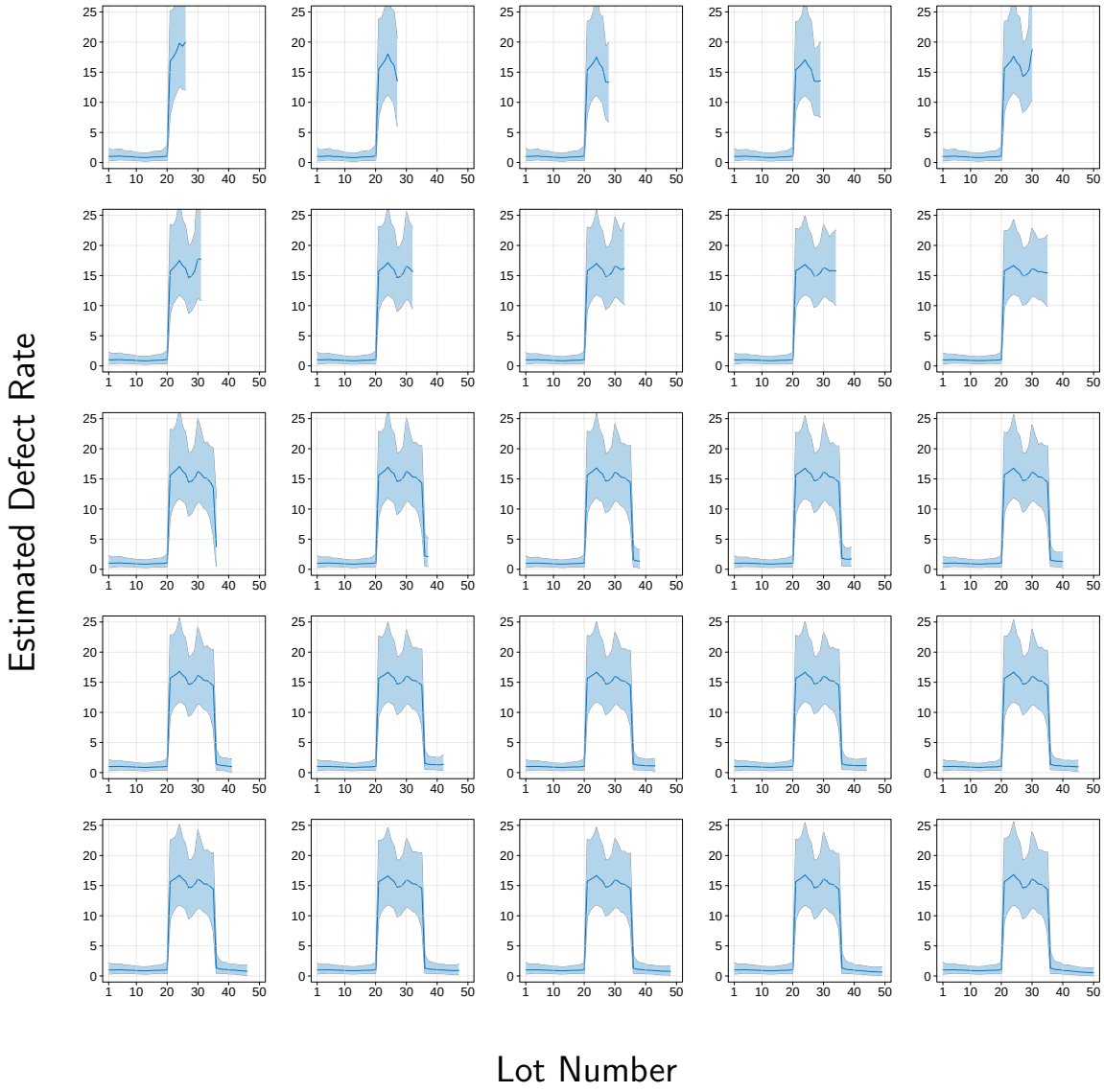


Figure 18: PART II, Abrupt deterioration and recovery: Evolving posterior distributions of θ at each time point based on the Bayesian state-space model and the samples reused from the curtailed frequentist run. Each panel represents one day, starting from day 26 (Lot 26) until day 50 (of a total of 50 lots). The x-axis represents the lots from day 1 to n for each iteration, and the y-axis represents the estimated defect rate.

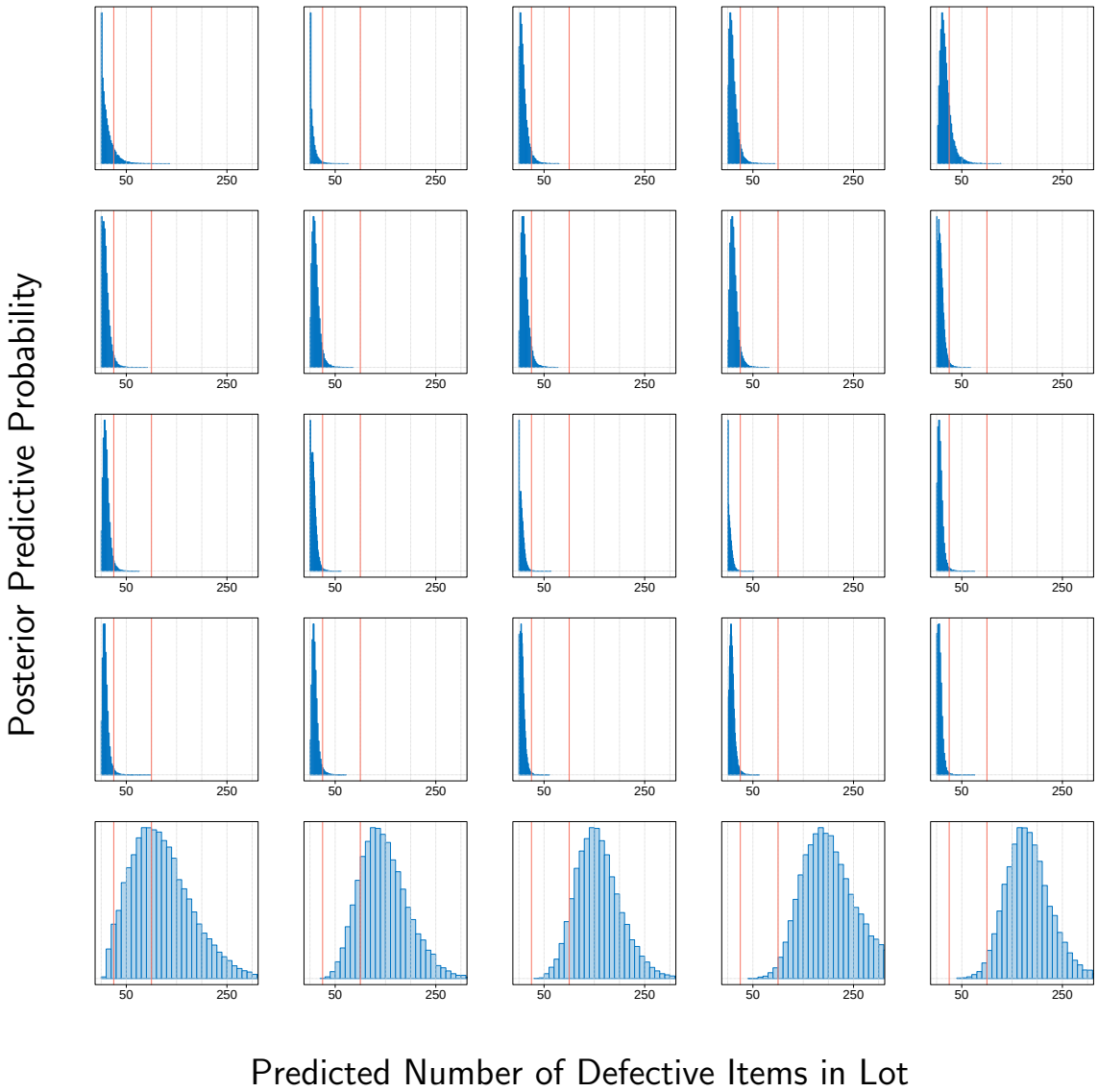


Figure 19: PART III, Abrupt deterioration and recovery: Predictive distributions for the total number of defective items in the first 25 (out of 50 lots) lots based on the Bayesian state-space model and the samples reused from the curtailed frequentist run.

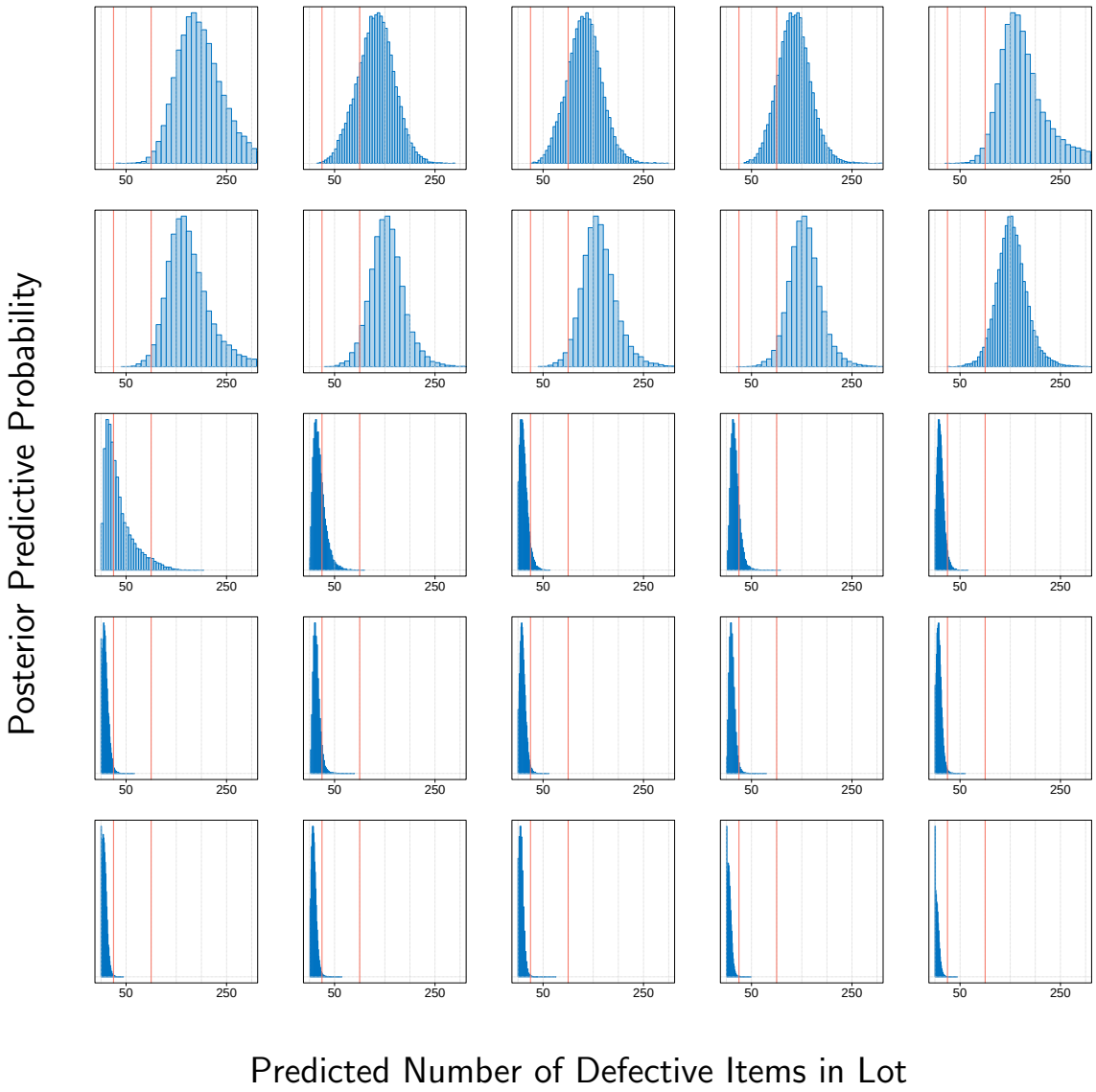


Figure 20: PART IV, Abrupt deterioration and recovery: Predictive distributions for the total number of defective items in the 26th to the 50th lot based on the Bayesian state-space model and the samples reused from the curtailed frequentist run.

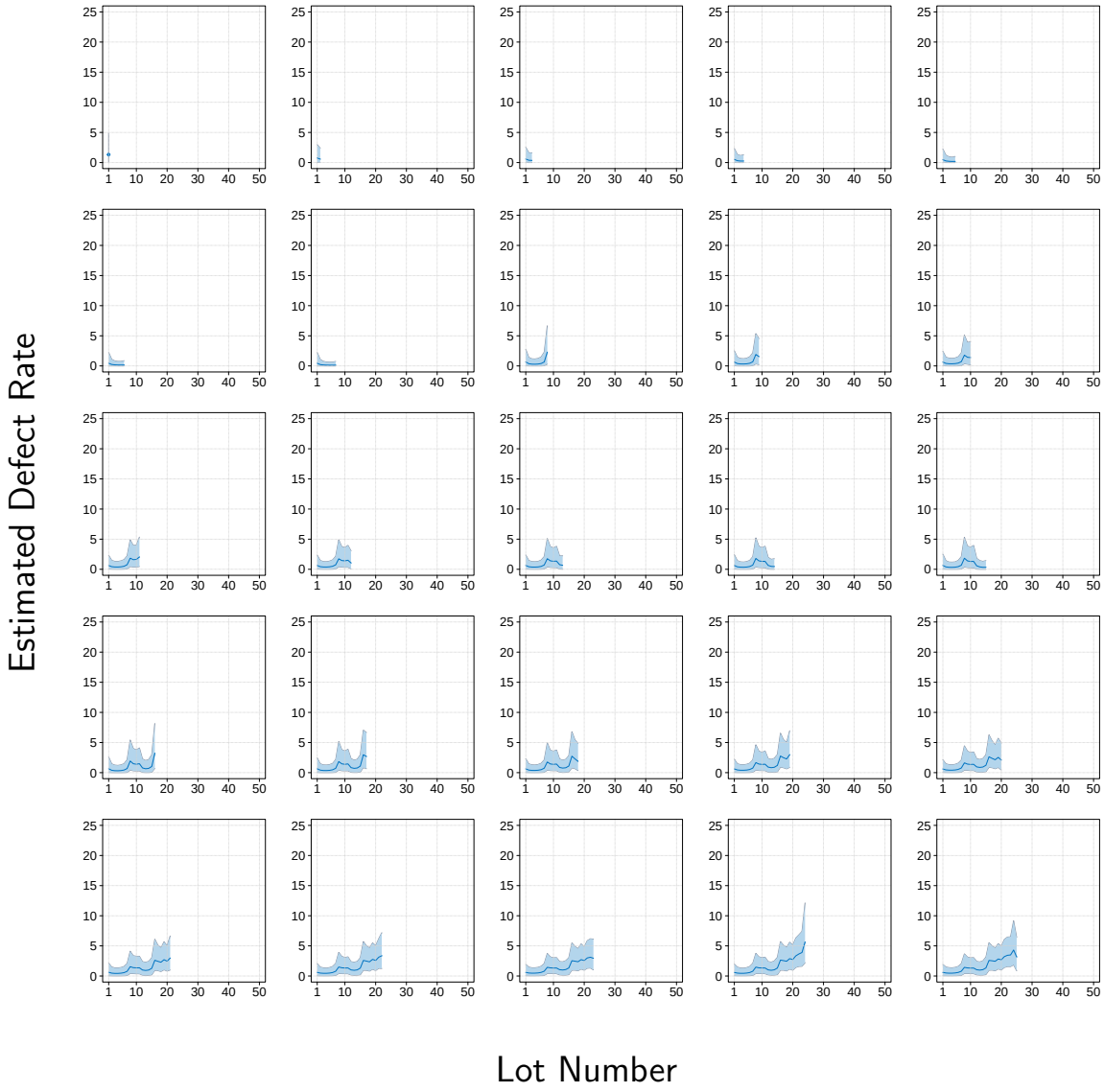


Figure 21: PART I, Gradual deterioration: Evolving posterior distributions of θ at each time point based on the Bayesian state-space model and the samples reused from the curtailed frequentist run. Each panel represents one day, starting from day 1 (Lot 1) until day 25 (of a total of 50 lots). The x-axis represents the lots from day 1 to n for each iteration, and the y-axis represents the estimated defect rate.

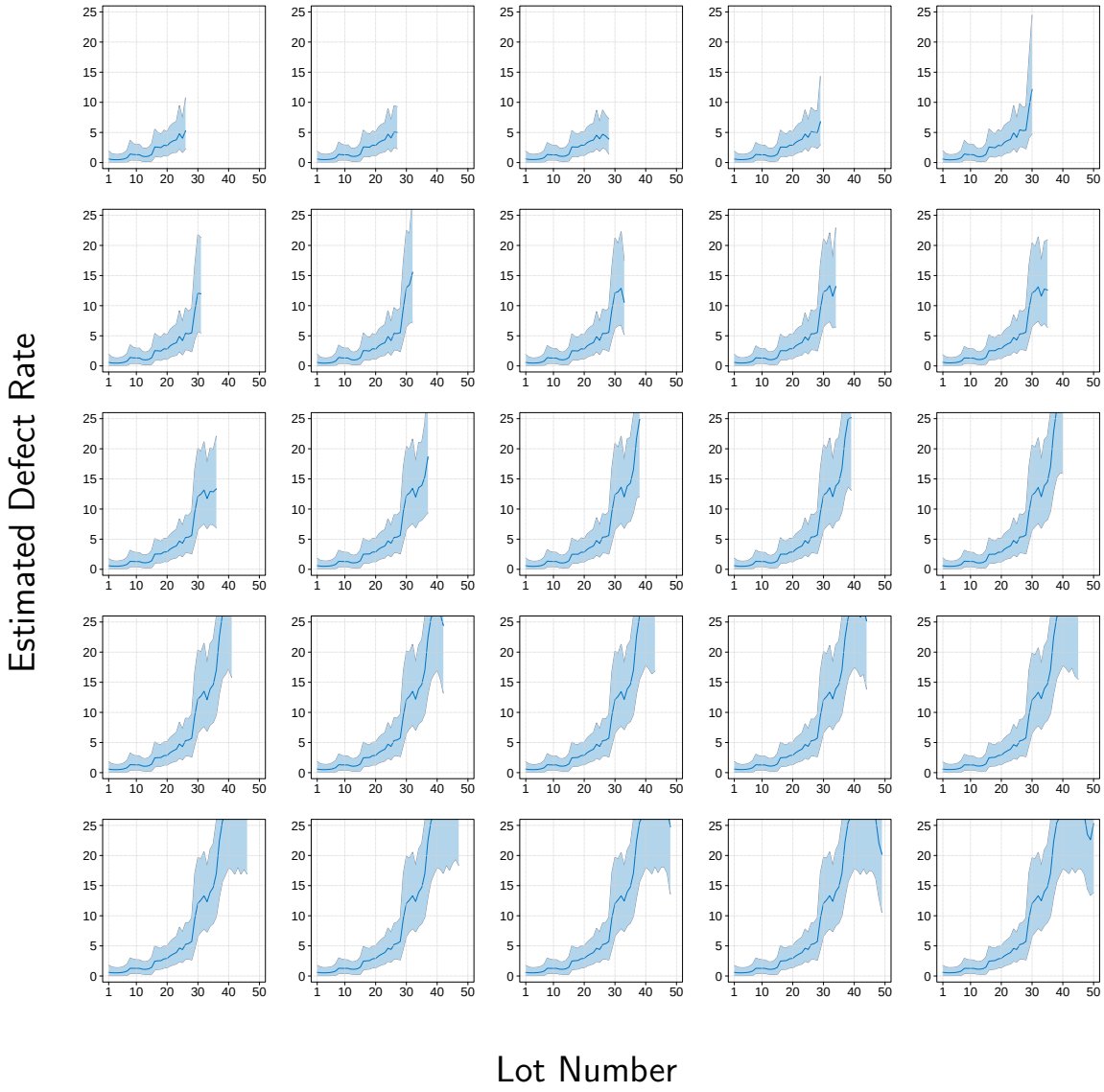


Figure 22: PART II, Gradual deterioration: Evolving posterior distributions of θ at each time point based on the Bayesian state-space model and the samples reused from the curtailed frequentist run. Each panel represents one day, starting from day 26 (Lot 26) until day 50 (of a total of 50 lots). The x-axis represents the lots from day 1 to n for each iteration, and the y-axis represents the estimated defect rate.

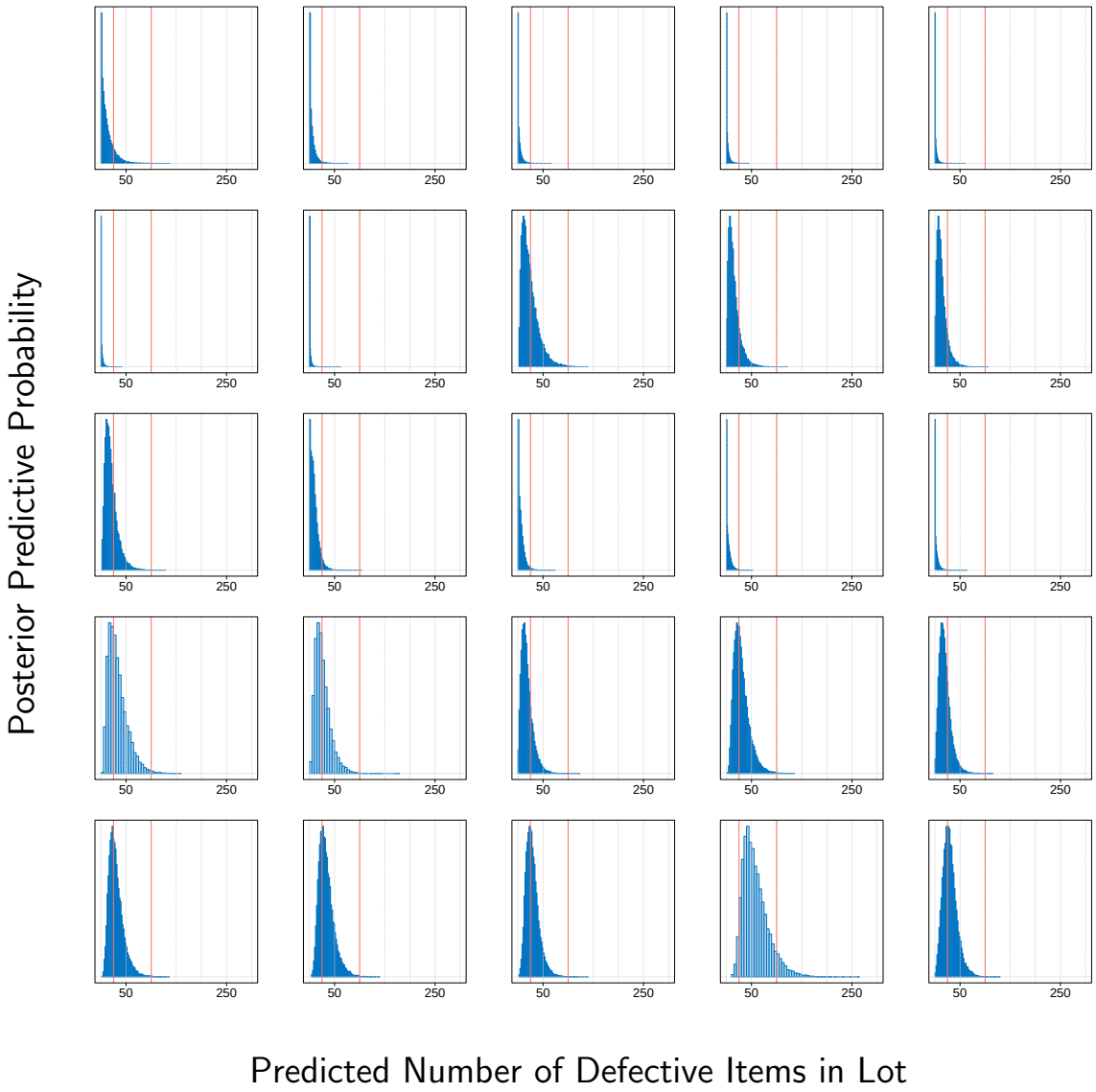


Figure 23: PART III, Gradual deterioration: Predictive distributions for the total number of defective items in the first 25 (out of 50 lots) lots based on the Bayesian state-space model and the samples reused from the curtailed frequentist run.

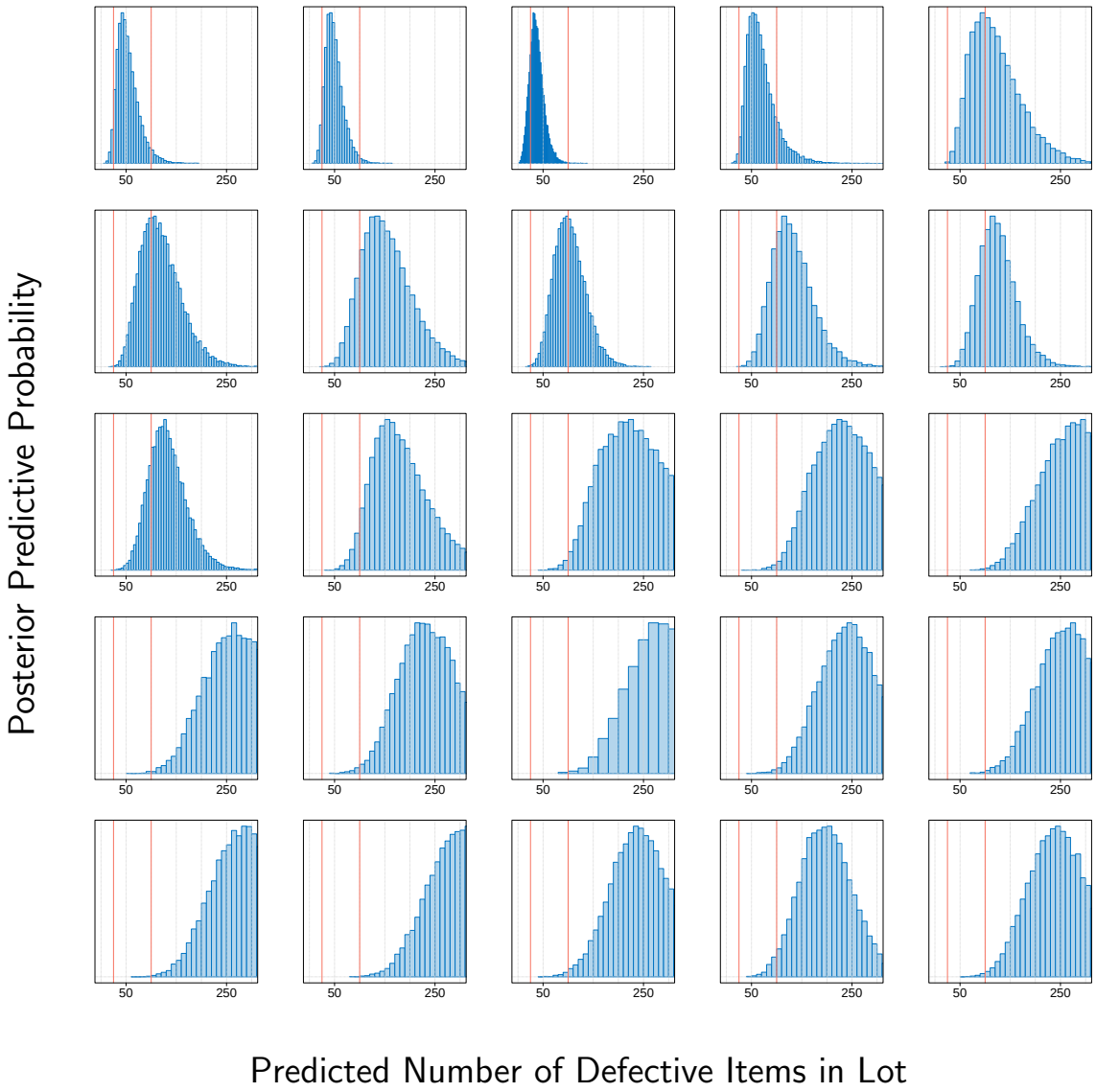


Figure 24: PART IV, Gradual deterioration: Predictive distributions for the total number of defective items in the 26th to the 50th lot based on the Bayesian state-space model and the samples reused from the curtailed frequentist run.

2 Sample Sizes Optional Stopping

Table 1: Bayesian optional-stopping results for scenario (a): Stable good quality.

Lots 1–25				Lots 26–50			
Lot	Items inspected	Defects observed	Decision	Lot	Items inspected	Defects observed	Decision
1	21	0	Accept	26	8	0	Accept
2	8	0	Accept	27	8	1	Accept
3	8	0	Accept	28	8	0	Accept
4	8	0	Accept	29	8	0	Accept
5	11	1	Accept	30	8	0	Accept
6	8	0	Accept	31	8	0	Accept
7	8	0	Accept	32	8	0	Accept
8	8	0	Accept	33	8	0	Accept
9	9	1	Accept	34	8	0	Accept
10	8	0	Accept	35	8	0	Accept
11	8	0	Accept	36	8	0	Accept
12	8	0	Accept	37	8	0	Accept
13	8	0	Accept	38	8	0	Accept
14	8	0	Accept	39	8	0	Accept
15	8	0	Accept	40	8	0	Accept
16	8	0	Accept	41	8	0	Accept
17	8	0	Accept	42	8	0	Accept
18	8	0	Accept	43	8	0	Accept
19	8	0	Accept	44	8	0	Accept
20	8	0	Accept	45	8	0	Accept
21	8	0	Accept	46	8	0	Accept
22	8	0	Accept	47	8	0	Accept
23	8	0	Accept	48	8	0	Accept
24	8	0	Accept	49	8	1	Accept
25	8	0	Accept	50	8	0	Accept

Table 2: Bayesian optional-stopping results for scenario (b): Stable intermediate quality.

Lots 1–25				Lots 26–50			
Lot	Items inspected	Defects observed	Decision	Lot	Items inspected	Defects observed	Decision
1	21	0	Accept	26	8	1	Reject
2	25	3	Reject	27	17	0	Accept
3	12	1	Reject	28	8	0	Accept
4	15	0	Accept	29	8	0	Accept
5	8	0	Accept	30	8	0	Accept
6	8	0	Accept	31	8	0	Accept
7	8	0	Accept	32	8	0	Accept
8	12	1	Accept	33	8	0	Accept
9	8	0	Accept	34	8	0	Accept
10	8	0	Accept	35	8	0	Accept
11	11	1	Accept	36	8	1	Accept
12	8	0	Accept	37	8	0	Accept
13	12	1	Accept	38	8	0	Accept
14	8	0	Accept	39	8	0	Accept
15	12	1	Accept	40	8	0	Accept
16	8	0	Accept	41	8	0	Accept
17	8	2	Reject	42	8	0	Accept
18	8	1	Reject	43	8	0	Accept
19	16	0	Accept	44	8	0	Accept
20	8	0	Accept	45	8	1	Accept
21	13	1	Accept	46	8	0	Accept
22	8	0	Accept	47	8	0	Accept
23	11	1	Accept	48	8	0	Accept
24	11	2	Reject	49	8	0	Accept
25	11	1	Reject	50	8	0	Accept

Table 3: Bayesian optional-stopping results for scenario (c): Stable poor quality.

Lots 1–25				Lots 26–50			
Lot	Items inspected	Defects observed	Decision	Lot	Items inspected	Defects observed	Decision
1	8	1	Reject	26	8	0	Reject
2	8	2	Reject	27	12	1	Reject
3	8	3	Reject	28	8	1	Reject
4	8	1	Reject	29	8	1	Reject
5	12	1	Reject	30	8	1	Reject
6	8	2	Reject	31	8	3	Reject
7	8	2	Reject	32	8	1	Reject
8	8	1	Reject	33	8	0	Reject
9	13	1	Reject	34	8	2	Reject
10	8	1	Reject	35	8	2	Reject
11	8	3	Reject	36	8	1	Reject
12	8	0	Reject	37	8	1	Reject
13	8	2	Reject	38	8	0	Reject
14	8	0	Reject	39	8	2	Reject
15	16	1	Reject	40	8	1	Reject
16	11	1	Reject	41	8	1	Reject
17	9	1	Reject	42	8	2	Reject
18	8	4	Reject	43	8	0	Reject
19	8	1	Reject	44	8	3	Reject
20	8	1	Reject	45	8	0	Reject
21	8	1	Reject	46	8	1	Reject
22	8	2	Reject	47	8	2	Reject
23	8	1	Reject	48	8	1	Reject
24	8	1	Reject	49	8	2	Reject
25	8	1	Reject	50	8	0	Reject

Table 4: Bayesian optional-stopping results for scenario (d): Abrupt, sustained deterioration.

Lots 1–25				Lots 26–50			
Lot	Items inspected	Defects observed	Decision	Lot	Items inspected	Defects observed	Decision
1	21	0	Accept	26	8	2	Reject
2	8	0	Accept	27	8	1	Reject
3	8	0	Accept	28	8	2	Reject
4	8	0	Accept	29	8	1	Reject
5	8	0	Accept	30	8	2	Reject
6	8	0	Accept	31	8	0	Reject
7	8	0	Accept	32	8	1	Reject
8	8	0	Accept	33	8	2	Reject
9	8	1	Accept	34	8	1	Reject
10	8	0	Accept	35	8	1	Reject
11	8	0	Accept	36	8	1	Reject
12	8	0	Accept	37	8	1	Reject
13	8	0	Accept	38	8	1	Reject
14	8	0	Accept	39	8	3	Reject
15	8	0	Accept	40	8	2	Reject
16	8	0	Accept	41	8	2	Reject
17	8	0	Accept	42	8	1	Reject
18	8	0	Accept	43	8	2	Reject
19	8	0	Accept	44	8	1	Reject
20	8	0	Accept	45	8	0	Reject
21	21	4	Reject	46	8	2	Reject
22	8	1	Reject	47	8	1	Reject
23	8	1	Reject	48	8	2	Reject
24	8	2	Reject	49	8	2	Reject
25	8	1	Reject	50	8	2	Reject

Table 5: Bayesian optional-stopping results for scenario (e): Abrupt deterioration and recovery.

Lots 1–25				Lots 26–50			
Lot	Items inspected	Defects observed	Decision	Lot	Items inspected	Defects observed	Decision
1	21	0	Accept	26	8	2	Reject
2	8	0	Accept	27	8	1	Reject
3	8	0	Accept	28	8	2	Reject
4	8	0	Accept	29	8	1	Reject
5	8	0	Accept	30	8	2	Reject
6	8	0	Accept	31	8	0	Reject
7	8	0	Accept	32	8	1	Reject
8	8	0	Accept	33	8	2	Reject
9	8	1	Accept	34	8	1	Reject
10	8	0	Accept	35	8	1	Reject
11	8	0	Accept	36	8	0	Reject
12	8	0	Accept	37	40	1	Accept
13	8	0	Accept	38	8	0	Accept
14	8	0	Accept	39	8	0	Accept
15	8	0	Accept	40	8	0	Accept
16	8	0	Accept	41	8	0	Accept
17	8	0	Accept	42	8	0	Accept
18	8	0	Accept	43	8	0	Accept
19	8	0	Accept	44	8	0	Accept
20	8	0	Accept	45	8	0	Accept
21	21	4	Reject	46	8	0	Accept
22	8	1	Reject	47	8	0	Accept
23	8	1	Reject	48	8	0	Accept
24	8	2	Reject	49	8	0	Accept
25	8	1	Reject	50	8	0	Accept

Table 6: Bayesian optional-stopping results for scenario (f): Gradual deterioration.

Lots 1–25				Lots 26–50			
Lot	Items inspected	Defects observed	Decision	Lot	Items inspected	Defects observed	Decision
1	21	0	Accept	26	8	1	Accept
2	8	0	Accept	27	8	0	Accept
3	8	0	Accept	28	11	1	Accept
4	8	0	Accept	29	15	1	Accept
5	8	0	Accept	30	8	2	Reject
6	8	0	Accept	31	8	1	Reject
7	8	0	Accept	32	8	2	Reject
8	8	0	Accept	33	8	0	Reject
9	8	0	Accept	34	8	1	Reject
10	8	0	Accept	35	8	1	Reject
11	8	1	Accept	36	9	1	Reject
12	8	0	Accept	37	8	1	Reject
13	8	0	Accept	38	8	3	Reject
14	8	0	Accept	39	8	3	Reject
15	8	0	Accept	40	8	2	Reject
16	8	0	Accept	41	8	1	Reject
17	8	0	Accept	42	8	3	Reject
18	8	0	Accept	43	8	2	Reject
19	8	0	Accept	44	8	1	Reject
20	8	1	Accept	45	8	3	Reject
21	8	0	Accept	46	8	2	Reject
22	8	0	Accept	47	8	1	Reject
23	8	1	Accept	48	8	2	Reject
24	8	0	Accept	49	8	1	Reject
25	8	0	Accept	50	8	2	Reject

3 Predictor Model Predictive Performance

Table 7: Predictive scores by lot for models without and with a predictor.

Lots 1–25					Lots 26–50				
	RPS		LogS			RPS		LogS	
Lot	Without	With	Without	With	Lot	Without	With	Without	With
1	—	—	—	—	26	5.492	4.195	4.093	3.692
2	6.857	4.792	3.926	3.777	27	9.308	12.038	4.344	4.548
3	7.624	7.301	4.042	4.008	28	5.295	4.011	3.826	3.559
4	7.503	7.097	4.279	4.242	29	8.766	9.888	4.523	4.523
5	5.022	5.236	3.945	3.956	30	5.295	3.376	4.040	3.564
6	12.232	9.258	4.277	4.082	31	9.060	4.185	4.314	3.773
7	22.942	15.566	5.422	4.770	32	23.513	20.834	5.276	5.244
8	5.042	6.453	3.928	4.002	33	11.096	10.062	4.619	4.412
9	15.199	8.270	4.917	4.456	34	13.074	3.190	4.422	3.462
10	6.262	4.307	4.293	3.776	35	6.406	5.486	4.215	4.007
11	20.423	10.329	5.068	4.341	36	14.754	4.258	4.465	3.378
12	34.851	14.109	5.999	4.662	37	5.433	4.520	4.025	3.673
13	18.475	2.646	4.725	3.269	38	4.451	2.803	3.866	3.391
14	15.042	5.811	4.643	3.699	39	8.331	4.355	3.983	3.598
15	4.318	3.978	3.764	3.694	40	10.068	4.927	4.443	3.890
16	3.811	3.727	3.693	3.626	41	9.739	5.794	4.435	4.098
17	20.254	24.154	5.070	6.267	42	5.904	6.469	4.148	4.095
18	4.341	5.737	3.669	3.864	43	12.749	13.480	4.553	4.460
19	3.707	6.521	3.643	3.971	44	19.651	4.756	4.805	3.808
20	13.896	18.669	4.806	5.065	45	6.083	5.312	4.113	3.846
21	7.252	5.452	4.093	3.969	46	22.933	6.817	5.231	4.156
22	4.874	5.321	3.864	3.778	47	15.372	6.939	4.594	3.847
23	17.570	10.844	4.616	4.279	48	28.268	6.491	5.473	4.216
24	7.909	12.999	4.387	4.520	49	20.907	9.551	5.012	4.118
25	32.913	11.983	5.702	4.588	50	12.789	26.428	4.394	5.727
Mean across lots 2–50						12.103	8.178	4.449	4.117

Note. Without and With refer to models without and with the predictor. RPS = ranked probability score. LogS = negative natural logarithm of the predictive probability assigned to the true full-lot defect count. Lower values indicate better performance for both scores. Each fit uses 78 inspected items from every lot up to and including the current lot. Lot 1 was not evaluated as the minimum lot number for the predictor model is 2. Means use all 49 evaluated lots and are calculated before rounding.